AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 10 · STUDENT LESSON

Human Classifier Simulation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 3: trace combined conditions at their boundaries.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Construct a classifier with a stated input format, selected rule and recorded output. Try the recall before opening notes.

Your learning target

I can construct a **classifier** with a stated **input** format, selected rule and recorded **output**.

I can trace valid, **boundary** and invalid **input**s through different paths.

I can distinguish an **input** check from **evidence** that a **prediction** is correct.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 10: Human **Classifier** Simulation

A **classifier** applies a learned or designed **boundary** to features.



THINKER MISSION

Train classmates on labeled creatures and compare their rules.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> A fitted rule predicts tall at a height of at least 6 cm and short otherwise. Predict at 5 and 6 cm. What should you check before applying it to a blank height?

This week's end goal

By the end, I can simulate a **classifier** with explicit requirements and missing-value behavior.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Human Classifier Simulation case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use human **classifier** simulation to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Your task is to build a paper **classifier** that another learner can operate without guessing your instructions. It will predict short or tall for imaginary towers.

Use the threshold of 6 centimetres selected after adding the height 5 short example in Week 13. Name this selected model Tower1.

Write an **input contract**: one height recorded as a whole number of centimetres from 1 through 10, including both endpoints. Require the unit cm to be shown. This is a classroom design choice about what the procedure accepts.

It is not a claim that every accepted height appeared in **fitting** or that towers outside this range cannot exist.

A complete **classifier** needs instructions for checking the **input**, applying Tower1 when the **input** is accepted, and recording the outcome. It must also say what happens when the **input** fails a check.

The threshold alone does not provide these surrounding instructions.

STUDENT LESSON · 3 OF 18

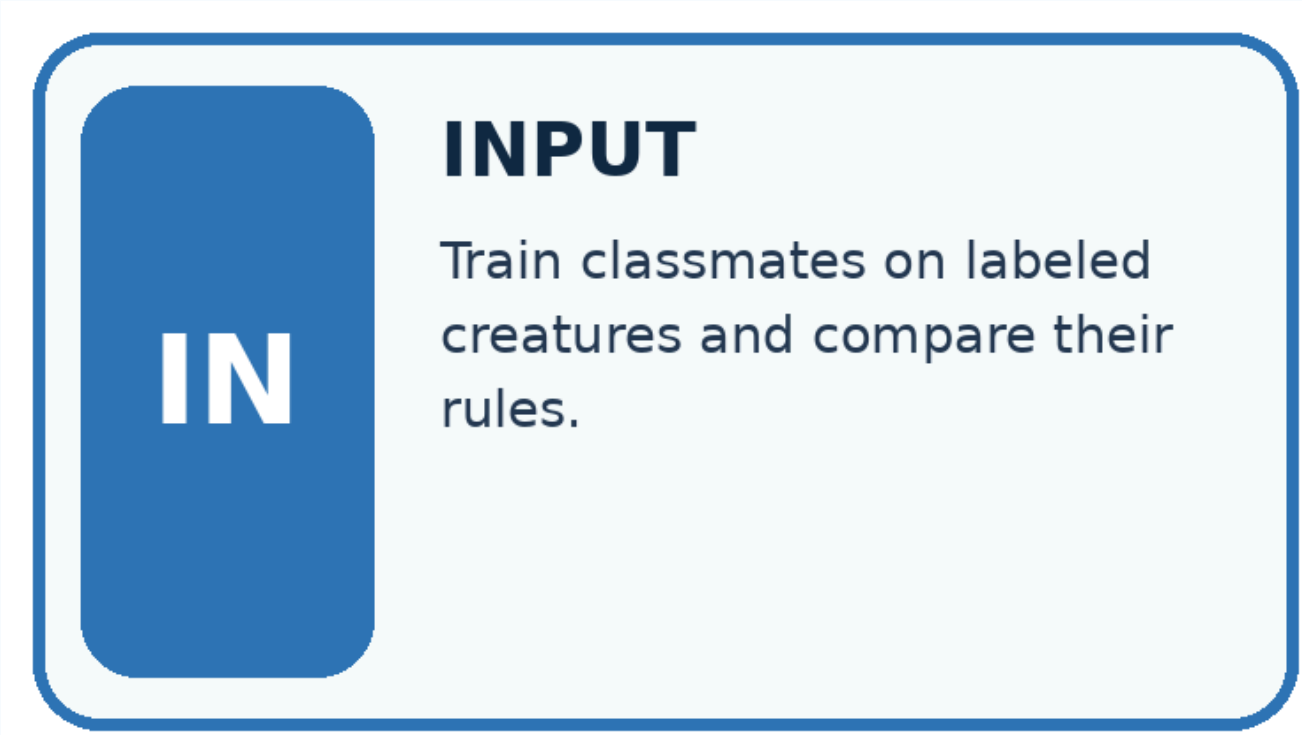
See how it works

> Follow the mechanism for Human Classifier Simulation. Connect each labelled input, deciding step and result.

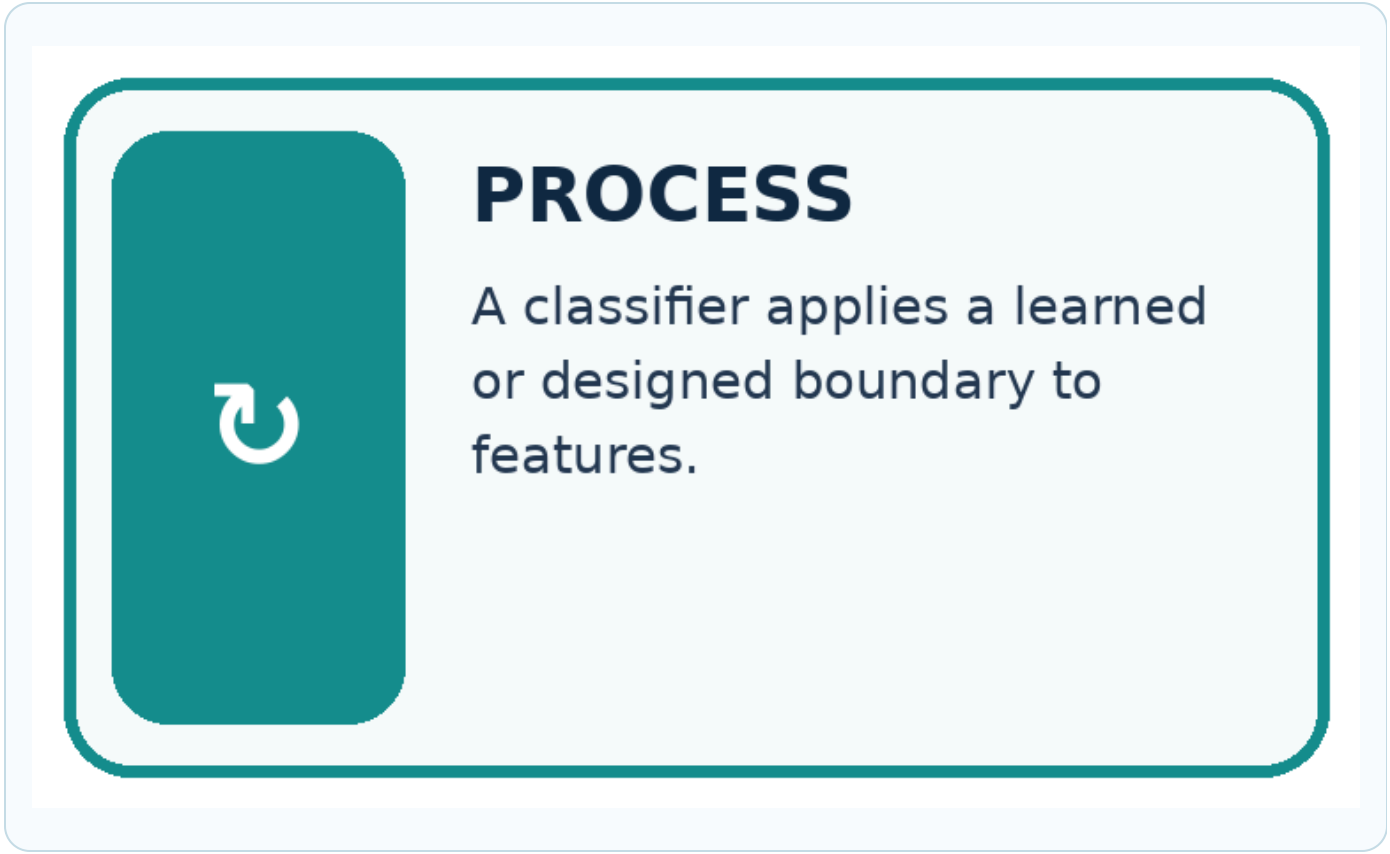
VISUAL MECHANISM

Input → Process → Output

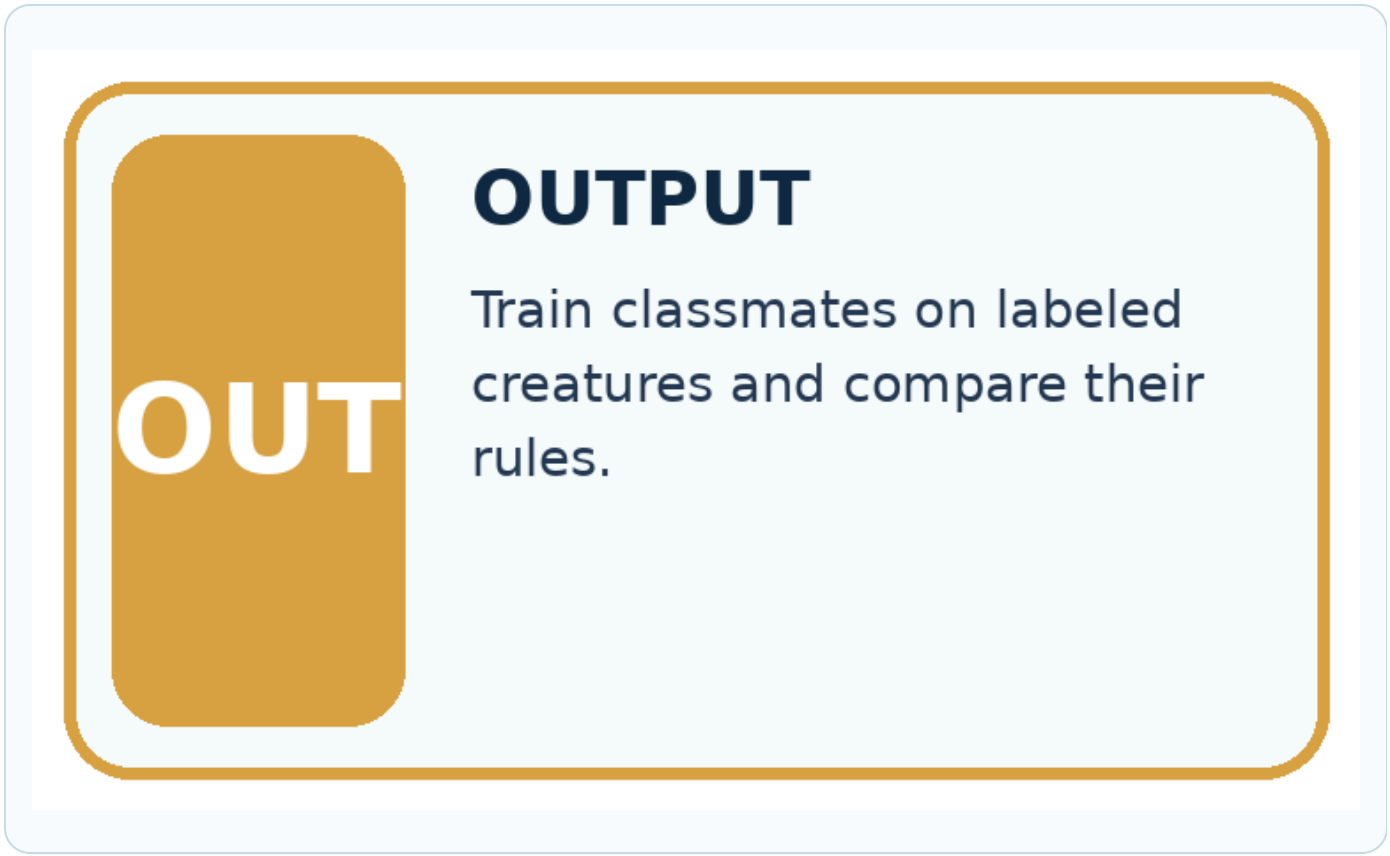
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
A classifier applies a learned or designed boundary to features.

↓


OUT
OUTPUT
Train classmates on labeled creatures and compare their rules.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Three records through the Tower1 classifier

INPUT RECORD	CHECK AND RULE	RECORDED OUTCOME
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

For every record, first check that a height is present and its stated unit is cm. Then check that it is a whole number from 1 through 10.

If any check fails, record **no prediction** and the reason, then stop **process**ing that record. Do not continue into the threshold comparison.

If all checks pass, compare the height with 6. At least 6 gives the **prediction** tall; otherwise give short. Record the original **input**, **model version** Tower1 and the **prediction**.

For a rejected record, record the original **input**, Tower1, **no prediction** and a reason. Tower1 identifies the intended model even when its threshold was not used.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

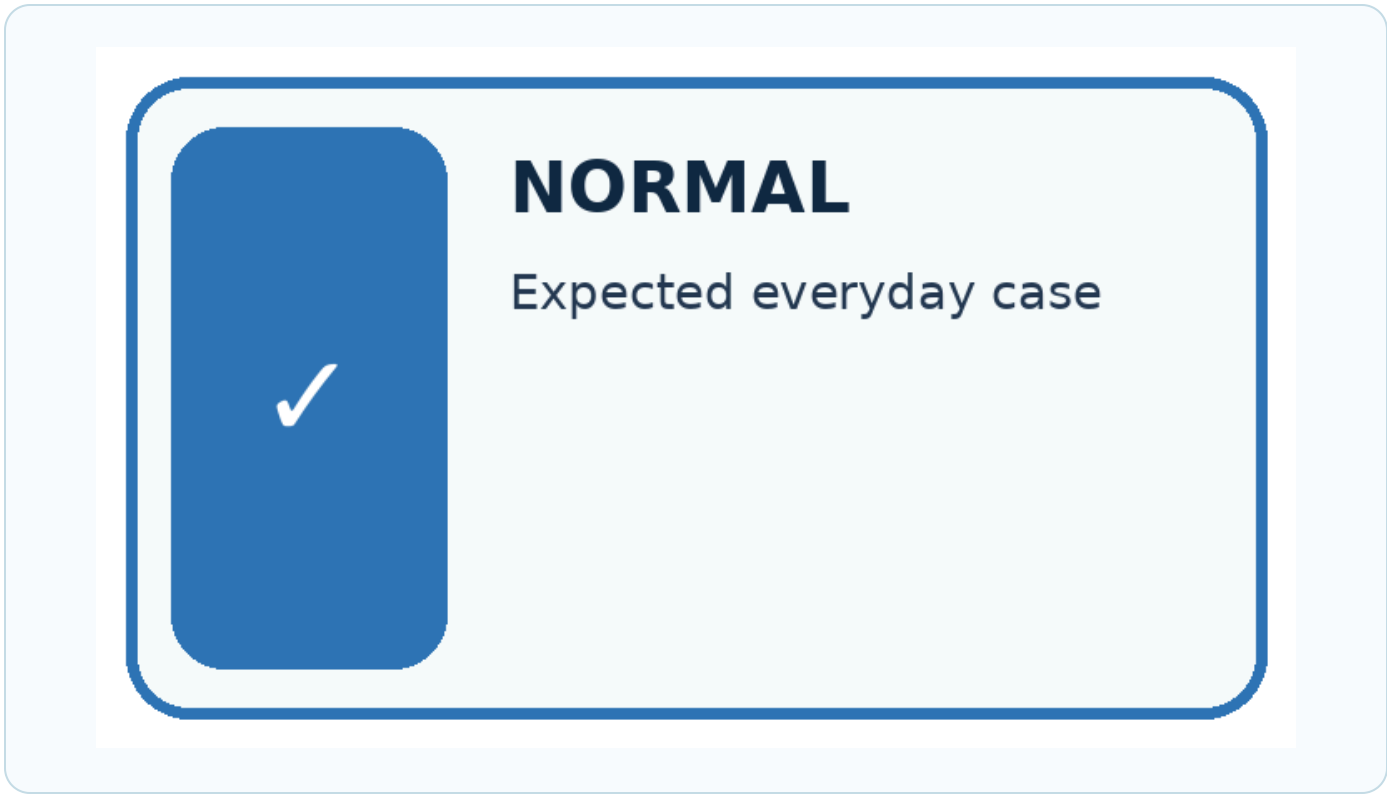
> Compare the supplied cases for Human Classifier Simulation. Identify the detail that changes the conclusion.

The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

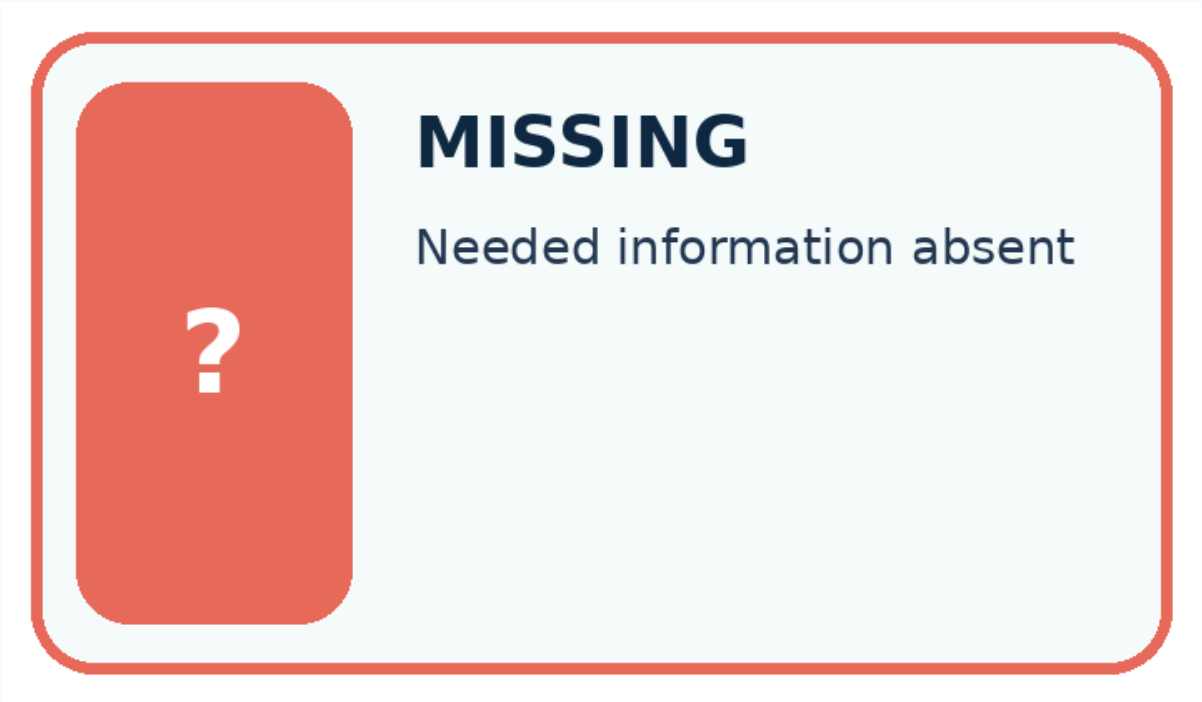
A system is not understood until it survives more than one comfortable example.


NORMAL
Expected everyday case


BOUNDARY
Near a decision edge

The normal case

NORMAL CASE: predict and name **evidence**. **BOUNDARY** CASE: predict and name **evidence**.





Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Supply M's pages as 101.

Predicted result

Prediction: M is now rejected rather than awaiting page review.

Why the result follows

Reason: The required field is known but fails the maximum-length condition.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Human Classifier Simulation to explain a concrete decision, rather than only memorising definitions.

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

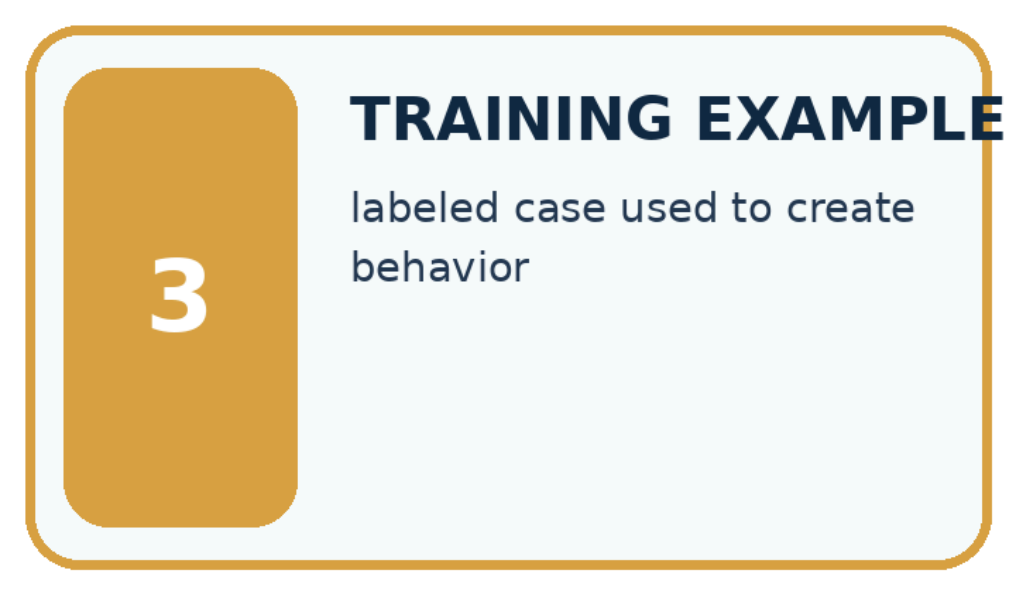
CLASSIFIER

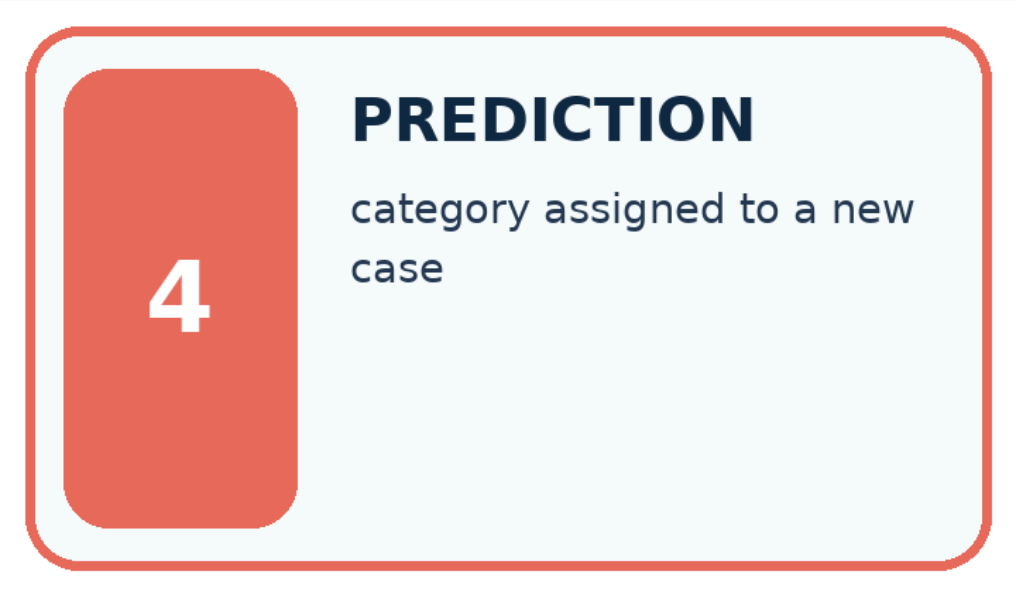
system that assigns a category

2

BOUNDARY

rule separating categories





Word	My own example	Not this / boundary
classifier	A rule or model that assigns an input to a category.	
boundary	A dividing condition between categories, actions or allowed behaviours.	
training example	An input and any associated label used to fit a model.	

Word	My own example	Not this / boundary
prediction	An estimated category, value or outcome that must be distinguished from established fact.	

Words used in the closer explanation

Word	Meaning
Input contract	The format and conditions an input must satisfy for a procedure to accept it
Validation	Checking whether an input satisfies the stated contract
No prediction	An explicit outcome when the classifier does not assign a category
Model version	A name that identifies the particular model used for a prediction

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STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Human Classifier Simulation, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 10 • WORKED EVIDENCE

1. Examples suggest: striped AND small → glim.
2. Learners infer a boundary from labeled cards.
3. New case: striped, small creature → predict glim.
4. Different inferred rules may match training yet disagree on a new case.

1. Examples suggest: striped AND small → glim.

2. Learners infer a **boundary** from labeled cards.

3. New case: striped, small creature → predict glim.

4. Different inferred rules may match **training** yet disagree on a new case.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Input	Contract result	Outcome
1 cm	Accepted at lower endpoint	Predict short
9 cm	Accepted	Predict tall
10 cm	Accepted at upper endpoint	Predict tall
0 cm	Outside 1 through 10	No prediction
60 mm	Unit is not cm	No prediction
4.5 cm	Not a whole number	No prediction

A valid 9 cm **input** receives a tall **prediction**, but the **input** check does not establish that the **prediction** matches its comparison label. **Validation** checks the contract.

Evaluation checks **prediction**s against appropriate **evidence**. Keep these questions separate.

Worked explanation · Human Classifier Simulation

Sixty millimetres equals six centimetres, but this version accepts cm records only. It does not contain a conversion step.

Ask for a correctly recorded cm measurement or deliberately design and check a conversion step in a new version. Do not silently treat the digits 60 as centimetres. A missing height must not become an invented zero either.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Human Classifier Simulation

A fictional librarian wants illustrated mystery books with at most 100 pages.

The facts we will use

Supplied facts

Written rule: choose only genre mystery AND illustrated True AND pages<=100.

Supplied facts

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing required fields request review.

AI ACADEMY / AI-2/10

Human Classifier Simulation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Read all conditions

Three requirements must pass, not just a high preference score.

2. Trace J

Mystery, illustrated and exactly 100 pages: choose.

3. Trace K and L

K fails illustration; L fails genre: do not choose.

4. Trace M

Missing pages means review rather than assuming it is short.

CHANGE ONE THING

Supply M's pages as 101.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read all conditions	Three requirements must pass, not just a high preference score.
2. Trace J	Mystery, illustrated and exactly 100 pages: choose.
3. Trace K and L	K fails illustration; L fails genre: do not choose.
4. Trace M	Missing pages means review rather than assuming it is short.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

This human simulation lets students inspect every decision without needing a trained model. Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check.

The order should not turn blank into zero or quietly ignore a required field.

What this example does not establish

Keep the stated limit

This declared rule checks eligibility, not whether an eligible book will actually be enjoyed.

Keep the stated limit

Read the labelled picture: Why are J and M initially treated differently even though both are illustrated mysteries?

Keep the stated limit

Change one thing: Supply M's pages as 101. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Human Classifier Simulation claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A classifier remembers the correct label for every possible future item. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

The **classifier** reports a **prediction** or **no prediction**. It does not decide what a person must do with a tower.

A separate classroom procedure might ask someone to remeasure a rejected **input** or compare a **prediction** with a recorded label. Write that action separately from the predicted category.

No prediction is not a third tower size. It records that this procedure did not assign short or tall.

Count and investigate such records when evaluating the whole system; hiding them could make a system look more useful than it is.

A classifier accepts 9 cm and predicts tall. A learner says the input passed, so the label is definitely right. Correct the claim. Then explain why reporting short for a blank height would violate the Tower1 procedure

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STUDENT LESSON · 8 OF 18

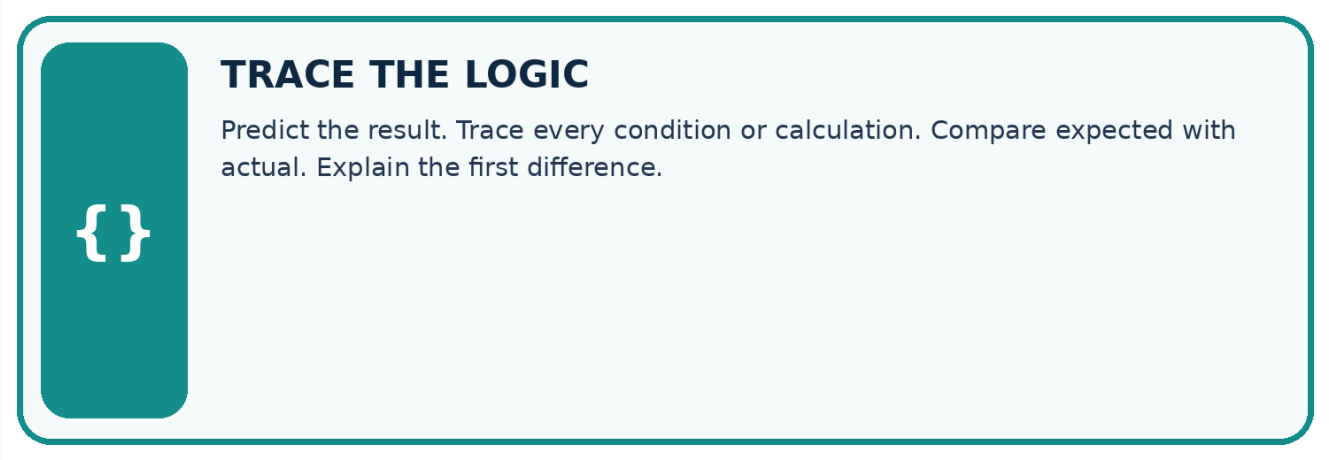
Practise together

Practise Human Classifier Simulation with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one **boundary** case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Write the procedure as numbered steps or connected boxes with labelled branches. Give it to a partner who follows only what you wrote.

The partner should record each check and stop point, including equality at the threshold and at both accepted range endpoints.

> **Construct Tower1 in your own words. Have a partner trace 1 cm, 6 cm, 10 cm, 11 cm and a blank height. Record each outcome and any instruction you had to clarify. Explain why the accepted 10 cm case still needs separate evidence of correctness**

A partner finding an unclear branch gives useful design feedback. Revise the instruction and retrace the affected examples.

Keep the model name and threshold visible so a change to the fitted rule is not confused with a correction to an instruction.

Practise with less help: Construct both outcome paths

Use workbook W2 for the connected example “Constructing a **classifier**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Constructing a **classifier**. Proposed explanation: “Any numeric **input** must receive a category”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Why are J and M initially treated differently even though both are illustrated mysteries?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

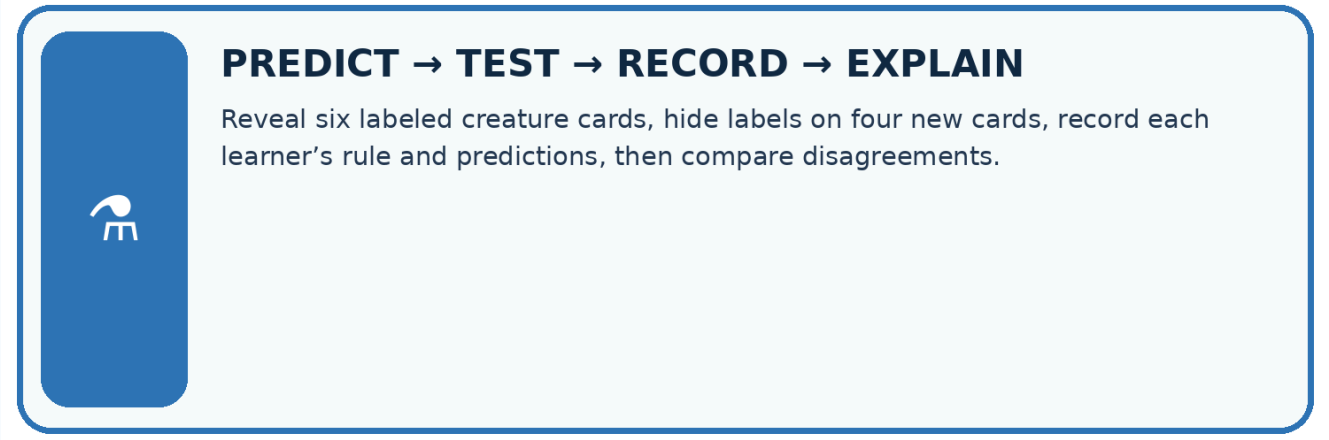
Run the investigation

> Investigate Human Classifier Simulation. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Reveal six labeled creature cards, hide labels on four new cards, record each learner’s rule and **prediction**s, then compare disagreements.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

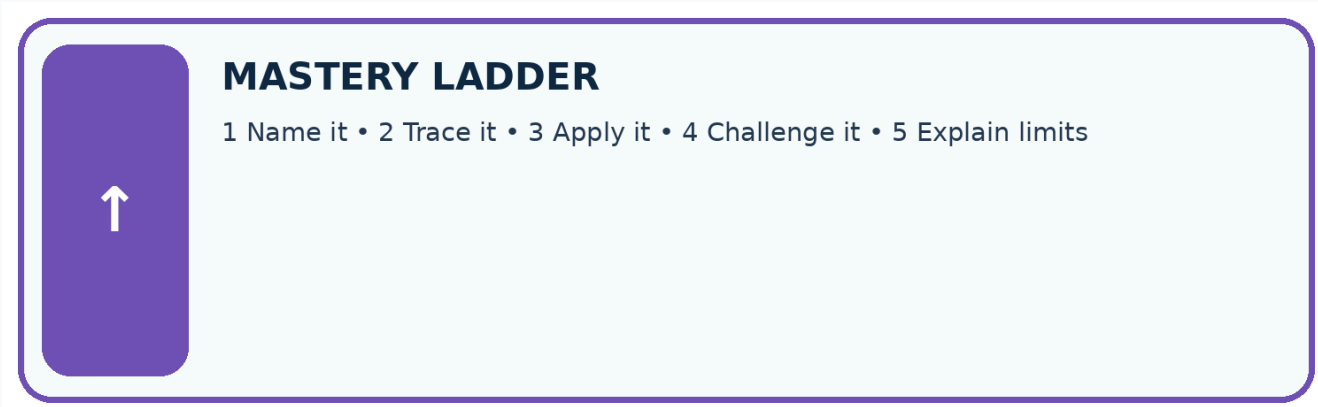
Build your understanding

Move from recall to tracing, diagnosis and transfer in Human Classifier Simulation. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a **boundary** case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

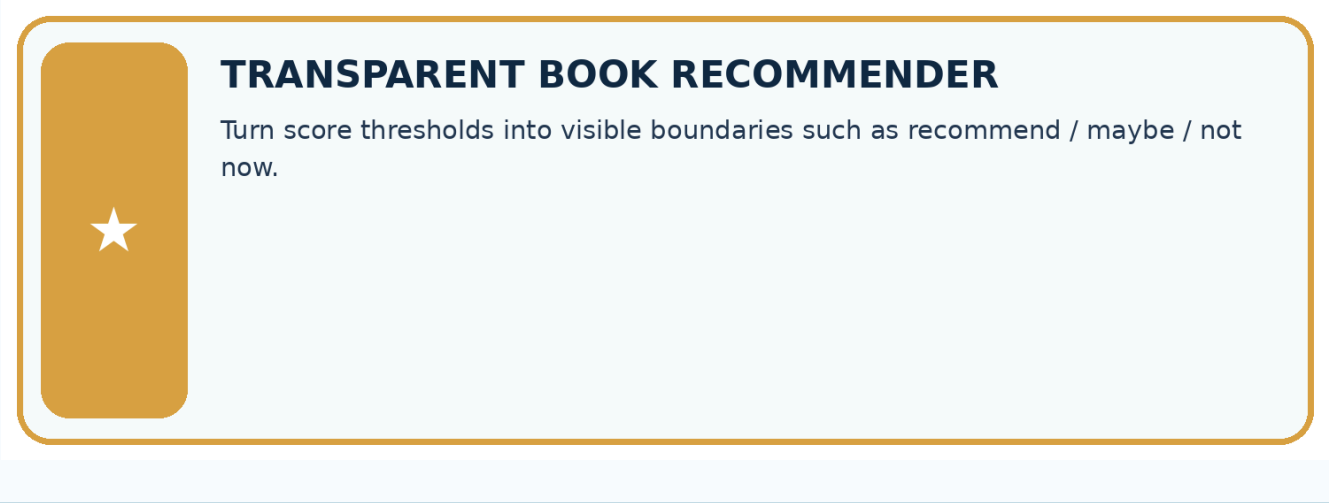
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Turn score thresholds into visible boundaries such as recommend / maybe / not now.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation.

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

✓

INDEPENDENT PROOF

Train classmates on labeled creatures and compare their rules.

Try a more demanding case

CHALLENGE Train classmates on labeled creatures and compare their rules.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a **boundary** or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Human Classifier Simulation case for supported practice. Repeated examples do not become fresh assessments.

Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Recommend if adventure AND pages<150; otherwise Other.

The question

Trace A=(adventure,120), B=(adventure,180), C=(science,90).

Worked explanation

A is Recommend. B fails length; C fails genre. Show both conditions.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Both conditions must pass

Evidence or stage	Value or result
Adventure, 120 pages	Genre yes, length yes → Recommend
Adventure, 180 pages	Genre yes, length no → Other
Science, 90 pages	Genre no, length yes → Other

Recommend if adventure AND pages<150; otherwise Other.

Trace A=(adventure,120), B=(adventure,180), C=(science,90).

A is Recommend. B fails length; C fails genre. Show both conditions.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Changed independent case

A **written rule** selects science books with fewer than 140 pages. Trace science/139, science/140, history/90 and missing-genre/90. Policy: missing required information means request it.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A human **classifier** uses sports topic means sports shelf. Classify a sports book and an unreadable cover. State what **evidence** is missing in the second case.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a **boundary** or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Turn score thresholds into visible boundaries such as recommend / maybe / not now.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Recommend a book only if genre is science AND pages<=150. Classify science/150, science/151, mystery/90 and missing-genre/80. State a missing-data policy.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Human Classifier Simulation, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional librarian wants illustrated mystery books with at most 100 pages.

By the end, I can	How I will show it
Construct a classifier with a stated input format, selected rule and recorded output	Explain the deciding step, show a trace or test, and state one limit.
Trace valid, boundary and invalid inputs through different paths	Explain the deciding step, show a trace or test, and state one limit.
Distinguish an input check from evidence that a prediction is correct	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Human Classifier Simulation without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Construct a **classifier** with a stated **input** format, selected rule and recorded **output**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional librarian wants illustrated mystery books with at most 100 pages.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Missing pages means review rather than assuming it is short.

1. Read all conditions

2. Trace M

3. Trace J

4. Trace K and L

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Trace M Missing pages means review rather than assuming it is short. This human simulation lets students inspect every decision without needing a trained model.

Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check. The order should not turn blank into zero or quietly ignore a required field.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Supply M's pages as 101. Which conclusion is supported by the supplied facts?

1. M is now rejected rather than awaiting page review.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: M is now rejected rather than awaiting page review. The required field is known but fails the maximum-length condition.

Repair a missing building block

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the **classifier** always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?.

Return to this week and explain what you changed.

Keep your completed work

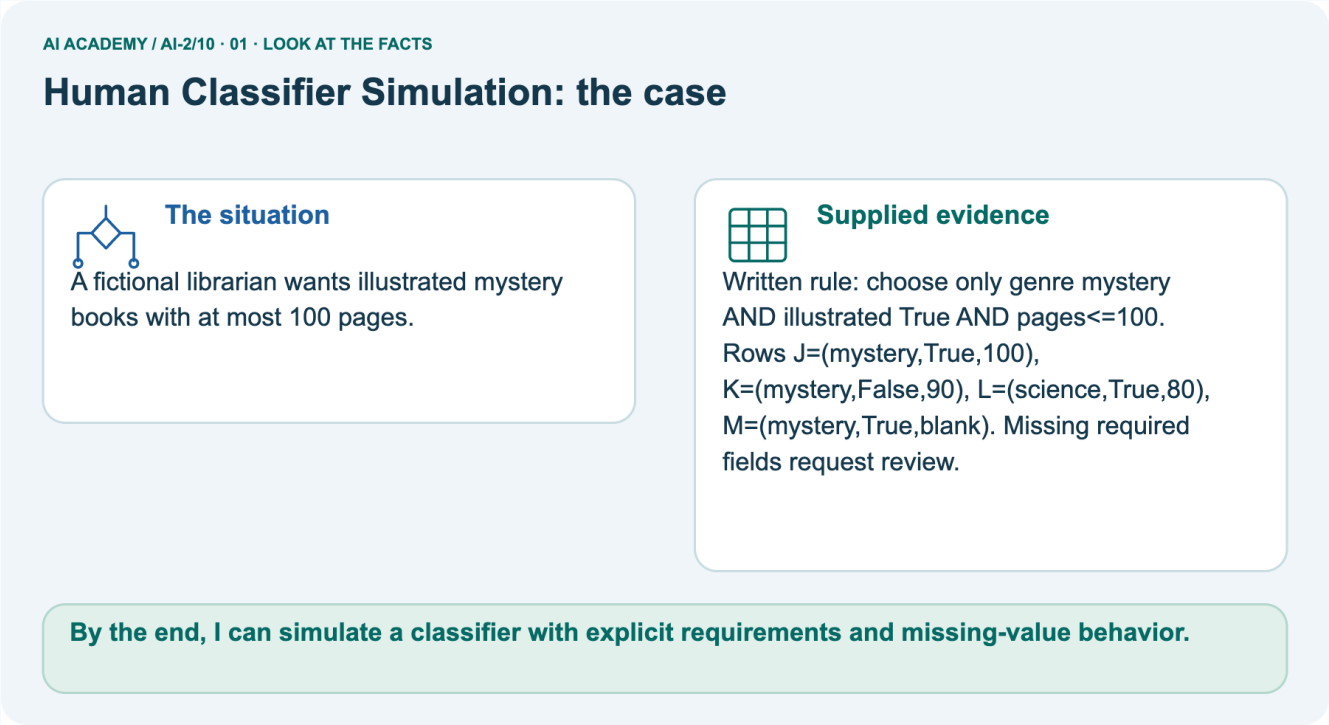
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Human Classifier Simulation graphic using its labels, deciding evidence and relevant limitation.

01. Human Classifier Simulation: the case



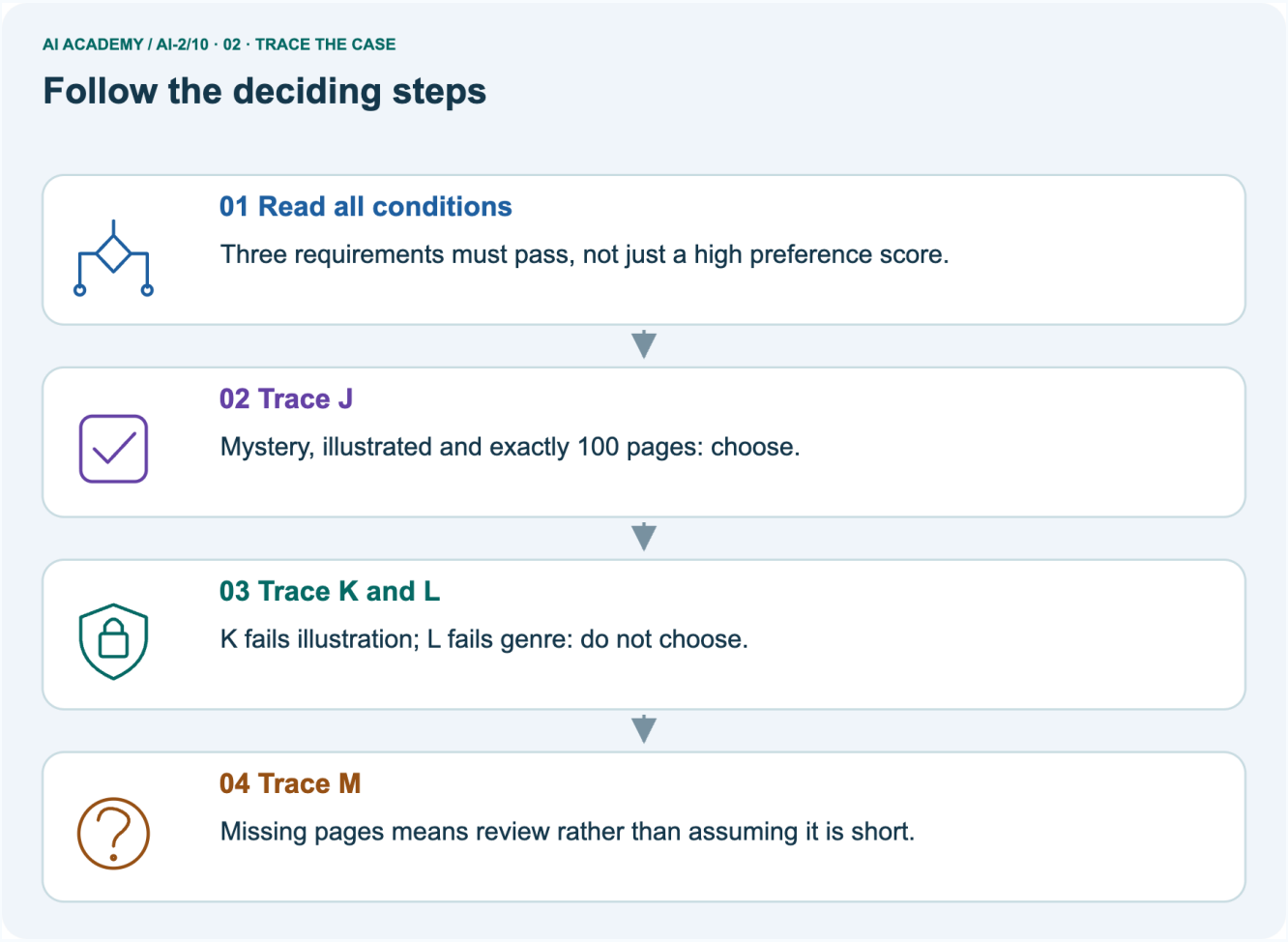
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional librarian wants illustrated mystery books with at most 100 pages. Supplied facts:
Written rule: choose only genre mystery AND illustrated True AND pages<=100.

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing required fields request review.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read all conditions: Three requirements must pass, not just a high preference score. → Trace J: Mystery, illustrated and exactly 100 pages: choose. → Trace K and L: K fails illustration; L fails genre: do not choose. → Trace M: Missing pages means review rather than assuming it is short.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Three records through the Tower1 classifier

AI ACADEMY / AI-2/10 · 03 · INSPECT THE LESSON MODEL

Three records through the Tower1 classifier

Input record	Check and rule	Recorded outcome
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

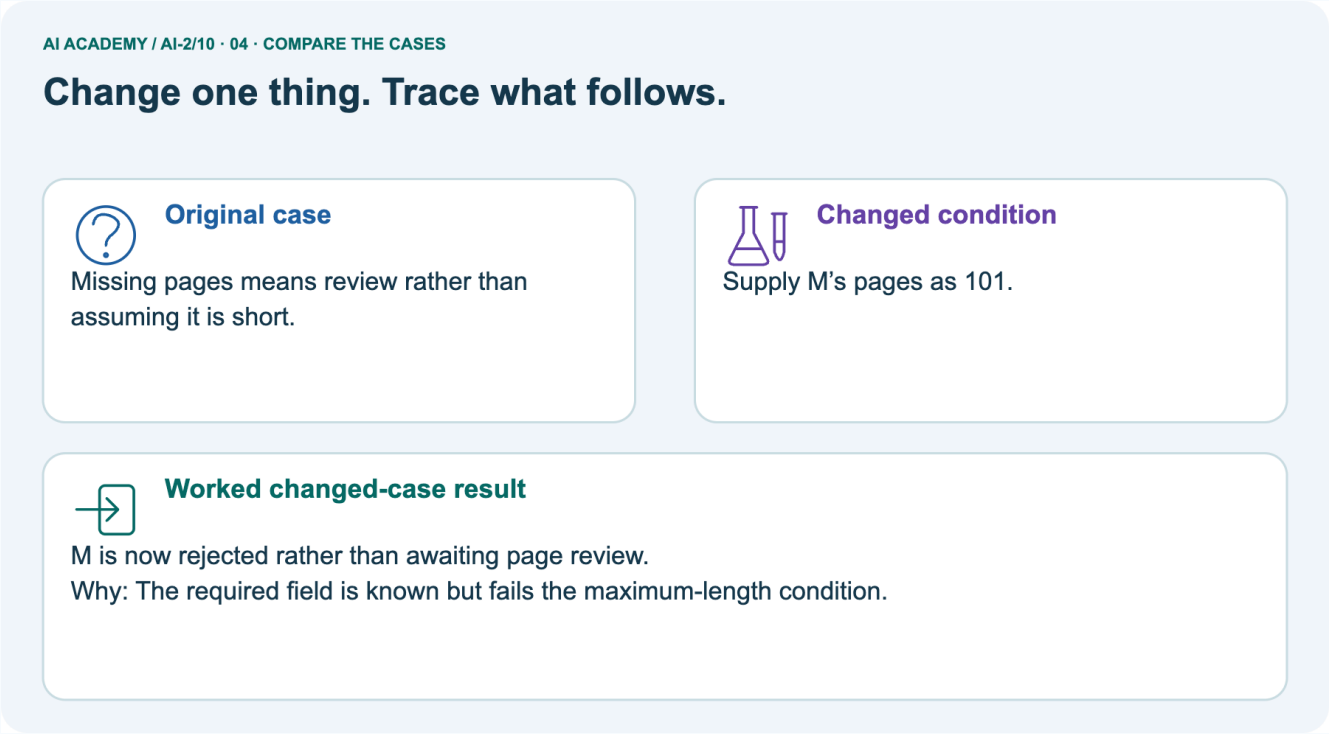
Follow the labelled picture with the matching explanation. Open the vector diagram

Three records through the Tower1 **classifier**. 5 cm | Accepted; 5 is below 6 | Tower1 predicts short. 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall.

Height blank | Missing height; stop before rule | Tower1; **no prediction**

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



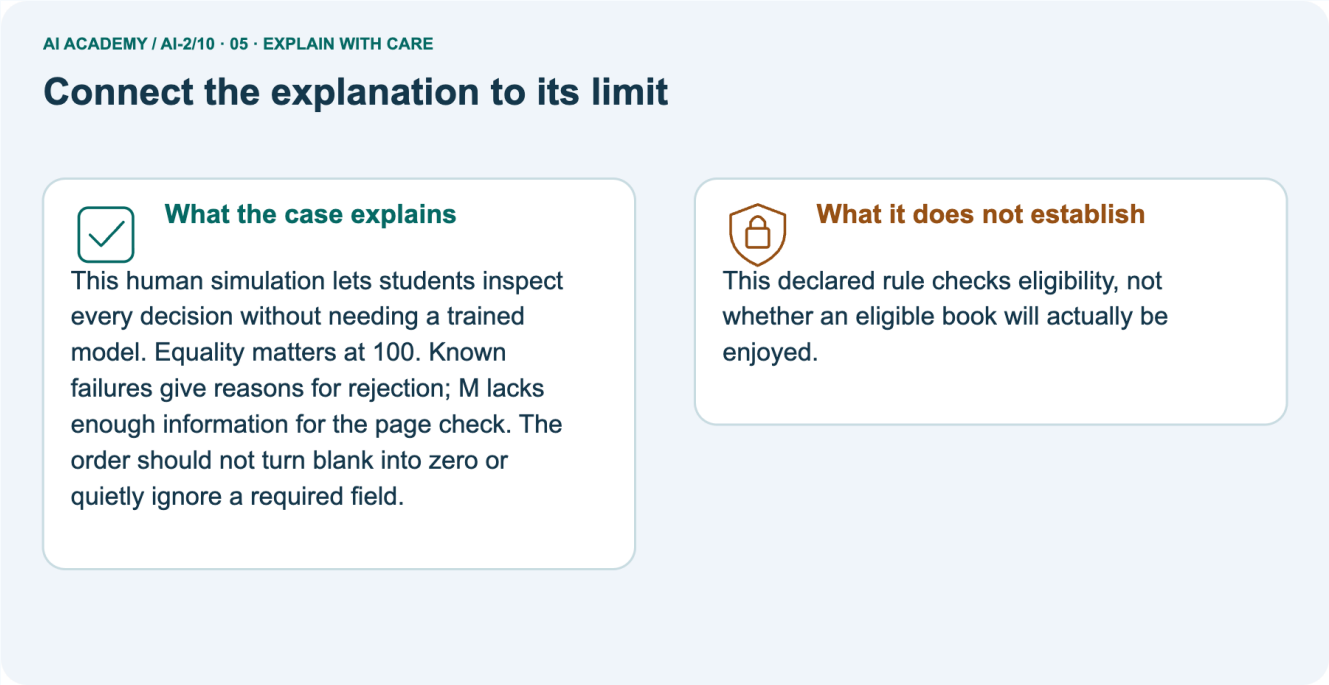
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Missing pages means review rather than assuming it is short. Changed condition: Supply M's pages as 101. Expected result: M is now rejected rather than awaiting page review.

Reason: The required field is known but fails the maximum-length condition.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



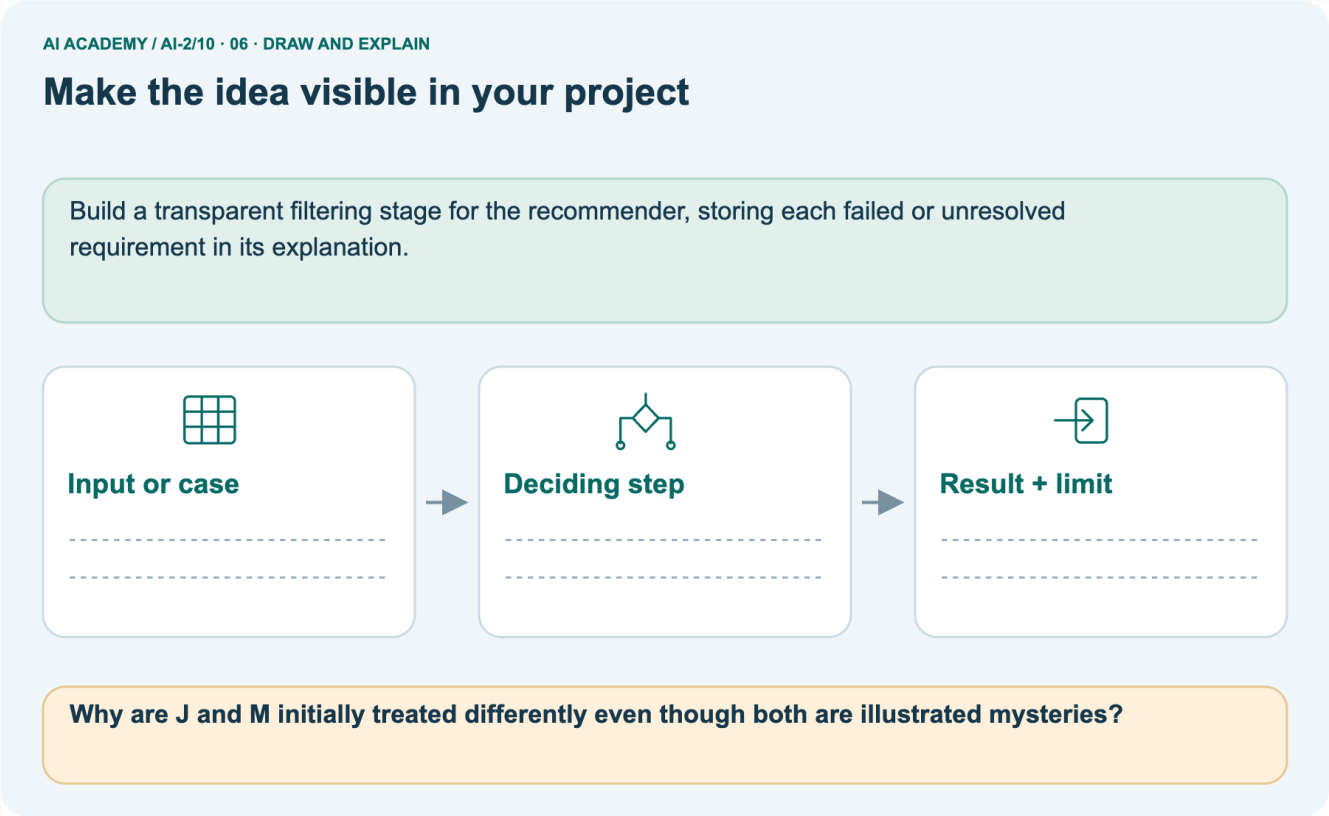
Follow the labelled picture with the matching explanation. Open the vector diagram

This human simulation lets students inspect every decision without needing a trained model. Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check.

The order should not turn blank into zero or quietly ignore a required field. Limit: This declared rule checks eligibility, not whether an eligible book will actually be enjoyed.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation. Draw your **input**, deciding step, result and limit.

Supported question: Why are J and M initially treated differently even though both are illustrated mysteries?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.